

Understanding Dockerfile Instructions and the Docker Image Build Process

Introduction

A **Dockerfile** is a text file that contains a set of instructions used to automatically build a Docker image. Each instruction creates a new image layer, making images reusable and efficient. Once an image is built, it can be used to create one or more containers.

What is a Docker Image?

A **Docker Image** is a read-only template that contains:

- Application code
- Runtime environment
- Libraries
- Dependencies
- Configuration files

An image is used to create Docker containers.

What is a Dockerfile?

A Dockerfile is a script containing instructions that Docker follows to build an image.

Common Dockerfile Instructions

1. FROM

Purpose

Specifies the base image from which the new image is built.

Syntax:

```
FROM ubuntu:22.04
```

Example:

```
FROM python:3.11
```

Without FROM, Docker cannot build an image.

2. RUN

Purpose

Executes commands during the image build process.

Used for:

- Installing software

- Updating packages
- Creating directories
- Downloading dependencies

Example:

`RUN apt-get update`

Example:

`RUN apt-get install -y nginx`

Each RUN instruction creates a new image layer.

3. COPY

Purpose

Copies files or directories from the host machine into the Docker image.

Syntax:

`COPY <source> <destination>`

Example:

`COPY . /app`

This copies all project files into the /app directory inside the image.

4. WORKDIR

Purpose

Sets the default working directory for subsequent instructions.

Example:

`WORKDIR /app`

After this instruction, commands such as RUN, CMD, and ENTRYPOINT execute inside /app.

5. EXPOSE

Purpose

Documents the network port on which the application listens.

Example:

`EXPOSE 80`

or

EXPOSE 8080

Note: EXPOSE does not publish the port; it simply documents it. Port publishing is done when running the container.

6. CMD

Purpose

Specifies the default command that runs when the container starts.

Example:

```
CMD ["python3", "app.py"]
```

Another example:

```
CMD ["nginx", "-g", "daemon off;"]
```

Only the last CMD instruction in a Dockerfile is used.

Docker Image Build Workflow

1. Write the Dockerfile.
2. Store application files in the project directory.
3. Run the docker build command.
4. Docker reads the Dockerfile from top to bottom.
5. Docker creates image layers for each instruction.
6. Docker stores the completed image locally.
7. Run the image to create a container.

Dockerfile

```
us-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-west-2&connType=standard&instanceId=i-0ff8df75f4ae3f168&osUser=ubuntu&sshPort=22&addressFamily=ipv4

AWS Search [Alt+S] United States (Oregon) Anukeerthana S (325137988786) Anukeerthana%20S

GNU nano 8.7.1 Dockerfile
FROM nginx:latest
WORKDIR /usr/share/nginx/html
COPY index.html .
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]

NG Help  NG Write Out  NG Where Is  NG Cut  NG Execute  NG Location  NG Undo  NG Set Mark  NG To Bracket  NG Previous
NX Exit  NR Read File  NR Replace  NR Paste  NJ Justify  NJ Go To Line  M-U Redo  M-A Copy  ME Where Was  ME Next

i-0ff8df75f4ae3f168 (Instance-01)
PublicIPs: 54.201.42.30 PrivateIPs: 172.31.45.167

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```

Docker Build Command

```
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker build -t docker-demo:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 3.072kB
Step 1/5 : FROM nginx:latest
latest: Pulling from library/nginx
e95a6c7ea7d4: Pulling fs layer
acr093e7a04f: Pulling fs layer
cd9307c9ecd8: Pulling fs layer
1645c1e06f46: Pulling fs layer
fcb6fd84b2a0: Pulling fs layer
df68ee7e7a00: Pulling fs layer
1b30016634d5: Pulling fs layer
cd9307c9ecd8: Download complete
fcb6fd84b2a0: Download complete
1645c1e06f46: Download complete
1b30016634d5: Download complete
acr093e7a04f: Download complete
jcf7d051b485: Download complete
e2c07e54e55a: Download complete
e95a6c7ea7d4: Download complete
df68ee7e7a00: Download complete
e95a6c7ea7d4: Pull complete
1b30016634d5: Pull complete
acr093e7a04f: Pull complete
df68ee7e7a00: Pull complete
cd9307c9ecd8: Pull complete
fcb6fd84b2a0: Pull complete
1645c1e06f46: Pull complete
Digest: sha256:ec4ed8b5299e5e90694af7750eb6dff2627317d30544d056b0371f8082f7bce
Status: Downloaded newer image for nginx:latest

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```

Image Tagging

```
Successfully tagged docker-demo:v1
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker tag docker-demo:v1 docker-demo:latest
root@ip-172-31-45-167:/home/ubuntu/docker_work#
```

```
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```

To list local images:

```
Successfully tagged docker-demo:v1
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker tag docker-demo:v1 docker-demo:latest
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:latest	258106a6db67	238MB	63.1MB	
docker-demo:v1	258106a6db67	238MB	63.1MB	
nginx:latest	ec4ed8b5299e	241MB	66MB	

```
root@ip-172-31-45-167:/home/ubuntu/docker_work#
```

```
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```

Running a Container

```
docker-demo:latest      258106a6db67      238MB      63.1MB
docker-demo:v1          258106a6db67      238MB      63.1MB
nginx:latest            ec4ed8b5299e      241MB      66MB
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker run -d -p 8080:80 --name docker-container docker-demo:v1
64cfb36e99f419293eadeec12733924e15f56451a3349b43a1963aff2ae78eea
root@ip-172-31-45-167:/home/ubuntu/docker_work#
```

i-Off8df75f4ae3f168 (Instance-01)

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Check Running Containers

```
nginx:latest            ec4ed8b5299e      241MB      66MB
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker run -d -p 8080:80 --name docker-container docker-demo:v1
64cfb36e99f419293eadeec12733924e15f56451a3349b43a1963aff2ae78eea
root@ip-172-31-45-167:/home/ubuntu/docker_work# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
64cfb36e99f4   docker-demo:v1   "/docker-entrypoint..."  34 seconds ago  Up 34 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  docker-container
root@ip-172-31-45-167:/home/ubuntu/docker_work#
```

i-Off8df75f4ae3f168 (Instance-01)

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